

# Prediction Intervals and Uncertainty for Short-Term Estimated WEC Power Forecasts

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Source notebook: [04\\_prediction\\_intervals\\_uncertainty.ipynb](#)

## 04 — Prediction Intervals and Uncertainty for Short-Term Estimated WEC Power Forecasts

This notebook extends the short-term point-forecasting baselines from [Notebook 03](#) by evaluating prediction intervals for the estimated WEC power signal.

Notebook 03 focused on point forecasts for the normalized estimated WEC power target, `wec_power_norm_estimated`, using a compact power-only autoregressive benchmark. The saved calibration and test predictions from that notebook provide the residual basis for the uncertainty analysis here.

The goal of this notebook is to evaluate whether simple empirical prediction intervals provide useful uncertainty estimates across forecast horizons, models, and higher-ramp periods. The main evaluation remains on the normalized WEC-power scale, while selected 250 kW-scale values are used later for physical interpretation of the generic WEC proxy.

### 1. Notebook 03 forecast outputs and analysis scope

Prediction intervals are built from the selected calibration and test predictions saved by [Notebook 03](#).

The calibration split is used to estimate residual-based interval rules. The test split is reserved for final prediction-interval evaluation. The main analysis focuses on Persistence, RollingMean, and Ridge, while RandomForest is retained only as a contrast model because it was slower and not competitive in the point-forecast benchmark.

Loaded forecast predictions: 102,396 rows

Loaded fold definitions: 85,299 rows

#### Notebook 04 input rows by horizon, model, and split

| horizon_steps | horizon_label | model        | split       | n_rows | first_target_time   | last_target_time    |
|---------------|---------------|--------------|-------------|--------|---------------------|---------------------|
| 1             | 30 min        | Persistence  | calibration | 3281   | 2025-04-18 16:00:00 | 2025-06-13 12:00:00 |
| 1             | 30 min        | Persistence  | test        | 3289   | 2025-05-08 03:00:00 | 2025-06-30 23:30:00 |
| 1             | 30 min        | RandomForest | calibration | 3281   | 2025-04-18 16:00:00 | 2025-06-13 12:00:00 |
| 1             | 30 min        | RandomForest | test        | 3289   | 2025-05-08 03:00:00 | 2025-06-30 23:30:00 |
| 1             | 30 min        | Ridge        | calibration | 3281   | 2025-04-18 16:00:00 | 2025-06-13 12:00:00 |
| 1             | 30 min        | Ridge        | test        | 3289   | 2025-05-08 03:00:00 | 2025-06-30 23:30:00 |
| 1             | 30 min        | RollingMean  | calibration | 3281   | 2025-04-18 16:00:00 | 2025-06-13 12:00:00 |

| horizon_steps | horizon_label | model        | split       | n_rows | first_target_time      | last_target_time       |
|---------------|---------------|--------------|-------------|--------|------------------------|------------------------|
| 1             | 30 min        | RollingMean  | test        | 3289   | 2025-05-08<br>03:00:00 | 2025-06-30<br>23:30:00 |
| 2             | 1 h           | Persistence  | calibration | 3250   | 2025-04-18<br>15:30:00 | 2025-06-13<br>13:30:00 |
| 2             | 1 h           | Persistence  | test        | 3258   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 2             | 1 h           | RandomForest | calibration | 3250   | 2025-04-18<br>15:30:00 | 2025-06-13<br>13:30:00 |
| 2             | 1 h           | RandomForest | test        | 3258   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 2             | 1 h           | Ridge        | calibration | 3250   | 2025-04-18<br>15:30:00 | 2025-06-13<br>13:30:00 |
| 2             | 1 h           | Ridge        | test        | 3258   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 2             | 1 h           | RollingMean  | calibration | 3250   | 2025-04-18<br>15:30:00 | 2025-06-13<br>13:30:00 |
| 2             | 1 h           | RollingMean  | test        | 3258   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 4             | 2 h           | Persistence  | calibration | 3189   | 2025-04-18<br>13:30:00 | 2025-06-13<br>16:30:00 |
| 4             | 2 h           | Persistence  | test        | 3195   | 2025-05-08<br>03:00:00 | 2025-06-30<br>23:30:00 |
| 4             | 2 h           | RandomForest | calibration | 3189   | 2025-04-18<br>13:30:00 | 2025-06-13<br>16:30:00 |
| 4             | 2 h           | RandomForest | test        | 3195   | 2025-05-08<br>03:00:00 | 2025-06-30<br>23:30:00 |
| 4             | 2 h           | Ridge        | calibration | 3189   | 2025-04-18<br>13:30:00 | 2025-06-13<br>16:30:00 |
| 4             | 2 h           | Ridge        | test        | 3195   | 2025-05-08<br>03:00:00 | 2025-06-30<br>23:30:00 |
| 4             | 2 h           | RollingMean  | calibration | 3189   | 2025-04-18<br>13:30:00 | 2025-06-13<br>16:30:00 |
| 4             | 2 h           | RollingMean  | test        | 3195   | 2025-05-08<br>03:00:00 | 2025-06-30<br>23:30:00 |
| 8             | 4 h           | Persistence  | calibration | 3065   | 2025-04-18<br>11:00:00 | 2025-06-14<br>15:00:00 |
| 8             | 4 h           | Persistence  | test        | 3072   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 8             | 4 h           | RandomForest | calibration | 3065   | 2025-04-18<br>11:00:00 | 2025-06-14<br>15:00:00 |
| 8             | 4 h           | RandomForest | test        | 3072   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |
| 8             | 4 h           | Ridge        | calibration | 3065   | 2025-04-18<br>11:00:00 | 2025-06-14<br>15:00:00 |
| 8             | 4 h           | Ridge        | test        | 3072   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |

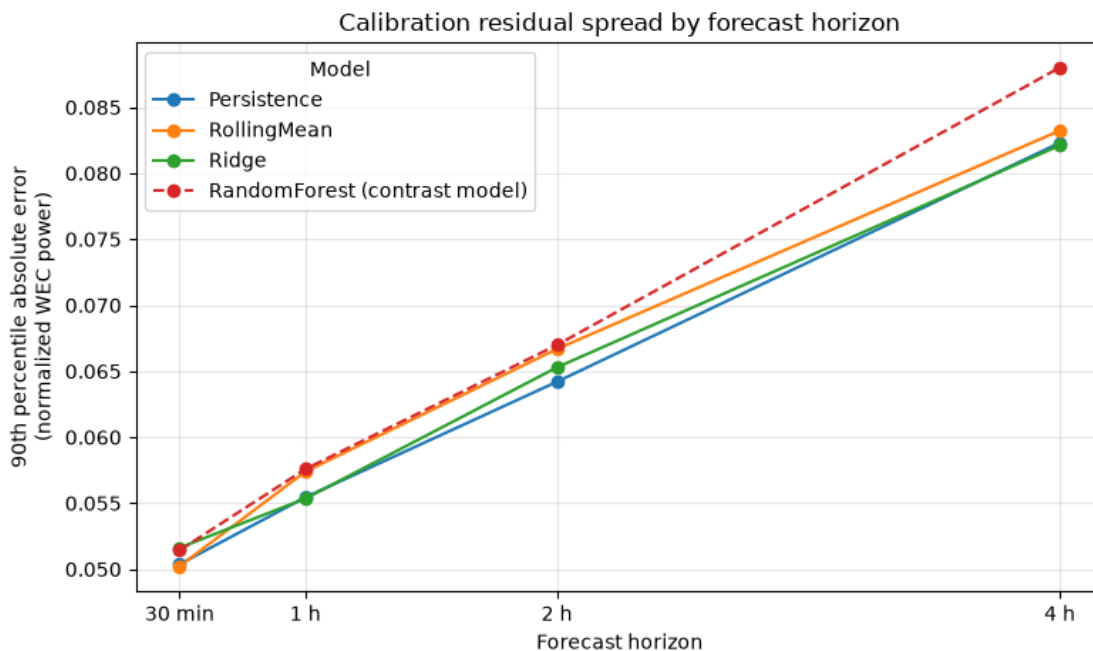
| horizon_steps | horizon_label | model       | split       | n_rows | first_target_time      | last_target_time       |
|---------------|---------------|-------------|-------------|--------|------------------------|------------------------|
| 8             | 4 h           | RollingMean | calibration | 3065   | 2025-04-18<br>11:00:00 | 2025-06-14<br>15:00:00 |
| 8             | 4 h           | RollingMean | test        | 3072   | 2025-05-08<br>03:30:00 | 2025-06-30<br>23:30:00 |

Saved input summary to: `../outputs/tables/notebook_04/interval_input_summary.csv`

## 2. Residual basis for interval calibration

Notebook 04 uses the calibration residuals saved by Notebook 03 as the empirical basis for prediction intervals. Before constructing intervals, the calibration residual spread is visualized by model and forecast horizon.

This provides a simple check of whether interval widths should depend on horizon and model choice.



Saved calibration residual spread figure to:

`../outputs/figures/notebook_04/calibration_residual_spread_by_horizon.png`

## 3. Build prediction intervals

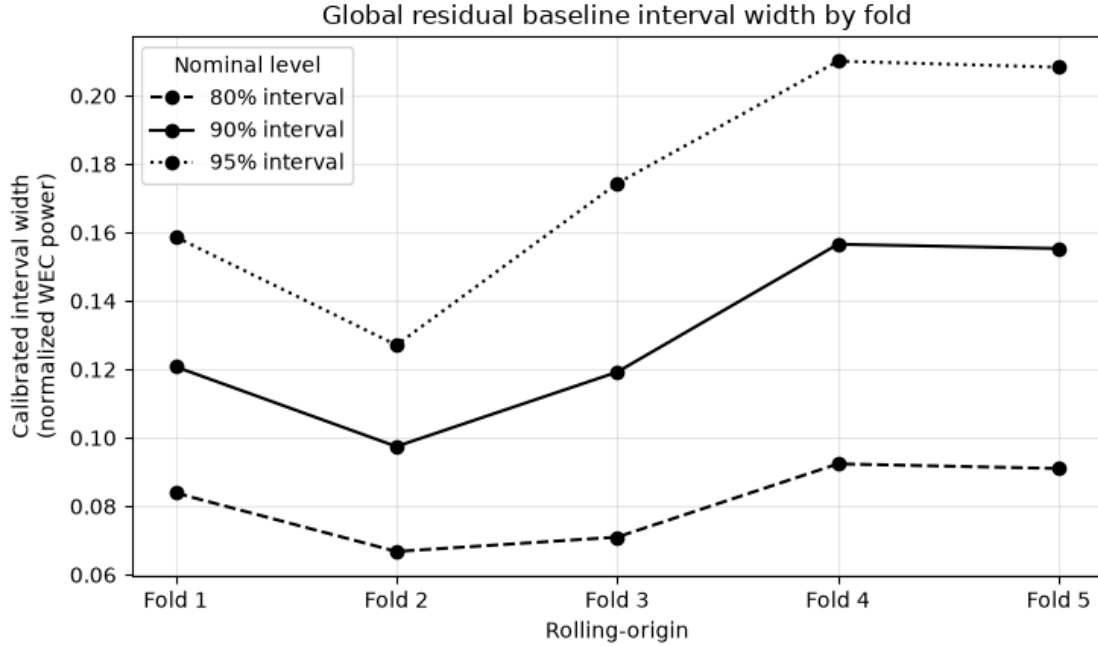
Prediction intervals are constructed from the calibration residuals saved by Notebook 03 and then applied to the test predictions.

The three core methods below are deliberately empirical and transparent. They reuse the same rolling-origin calibration/test structure as the point-forecasting notebook: calibration rows are used to estimate interval rules, while test rows are used only for final interval evaluation.

### 3.1 Global residual baseline

This baseline pools calibration residuals within each fold and interval level, then applies the same residual interval rule across models and horizons.

It is intentionally simple. Because it ignores model- and horizon-specific residual spread, it may under-cover harder horizons or produce unnecessarily wide intervals for easier horizons.

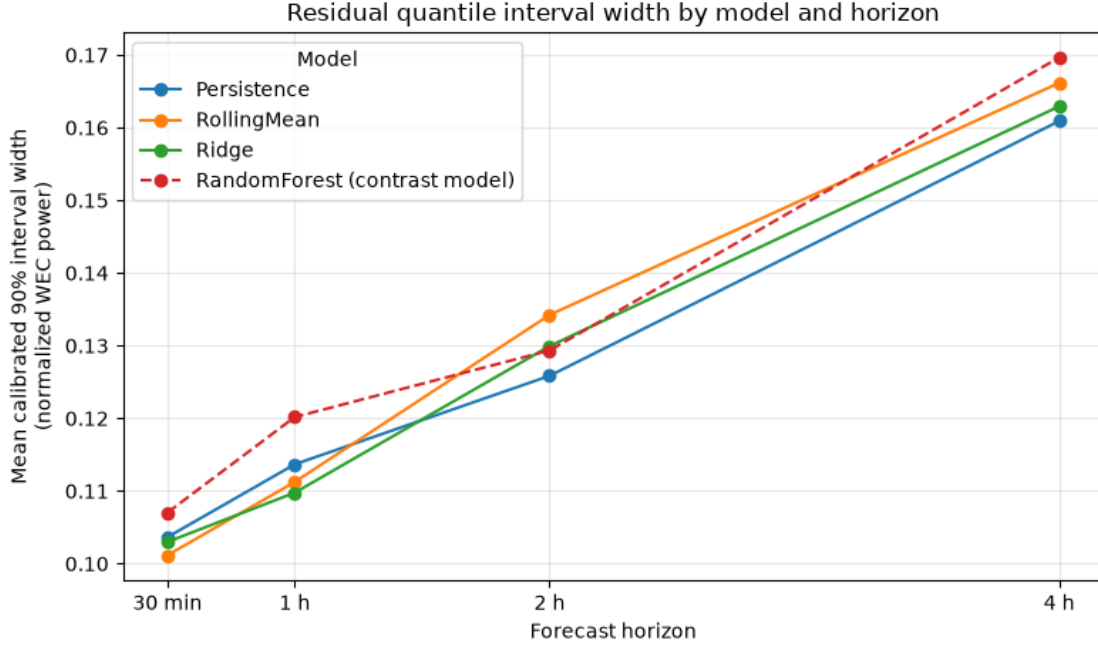


The global residual baseline gives wider intervals as the nominal coverage level increases, as expected. The calibrated widths also vary across rolling-origin folds, showing that the residual distribution changes over time. Because this baseline uses one pooled residual rule per fold, it does not yet adapt to the different uncertainty levels observed across forecast horizons or models.

### 3.2 Model- and horizon-specific residual quantile intervals

This method estimates lower and upper residual quantiles separately for each fold, model, forecast horizon, and interval level.

Unlike the global residual baseline, it allows interval width and asymmetry to change with both model behaviour and forecast horizon.



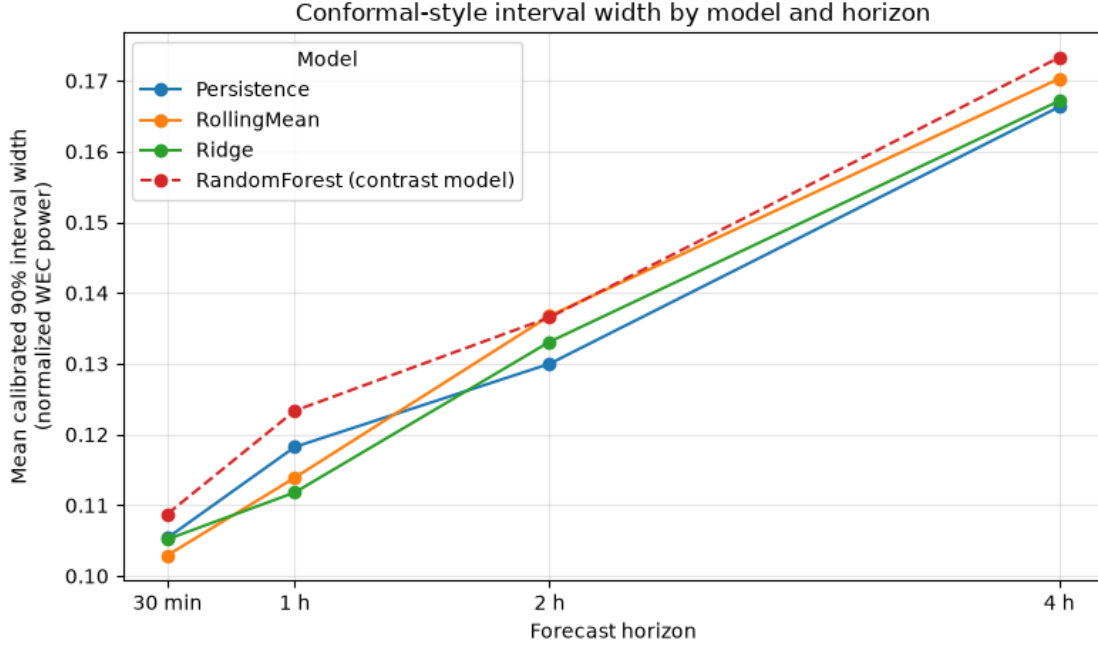
The model- and horizon-specific residual quantile intervals widen with forecast horizon, reflecting the increasing residual spread seen in the point-forecast diagnostics. The three main models have similar calibrated widths, while RandomForest is kept only as a contrast model. Compared with the global residual baseline, this method can adapt interval width to each model and forecast horizon instead of applying one pooled residual rule across all forecasts.

### 3.3 Conformal-style absolute-error intervals

This method uses the calibration absolute errors to build symmetric prediction intervals around each point forecast.

For each fold, model, forecast horizon, and interval level, it estimates a calibrated absolute-error quantile from the calibration split, then applies that value as a lower and upper correction around the test prediction.

This method is described as **conformal-style** rather than fully conformal. It follows the empirical calibration idea, but the time-series setting and earlier model-selection use of the calibration block mean that the intervals should be interpreted as practical calibrated intervals rather than strict formal guarantees.



The conformal-style intervals show the same horizon-dependent widening pattern as the residual quantile intervals. This is expected because both methods are calibrated from the same fold-specific residual behaviour, but the conformal-style method uses absolute errors and therefore produces symmetric intervals around the point forecast.

The main models again have similar calibrated widths, while RandomForest remains slightly wider and is kept only as a contrast model. These intervals are practical empirical intervals rather than formal guarantees, but they provide a simple calibrated uncertainty baseline for final test-set evaluation.

### 3.4 KDE residual-distribution intervals

This optional method estimates a smooth residual distribution from the calibration residuals for each fold, model, horizon, and interval level.

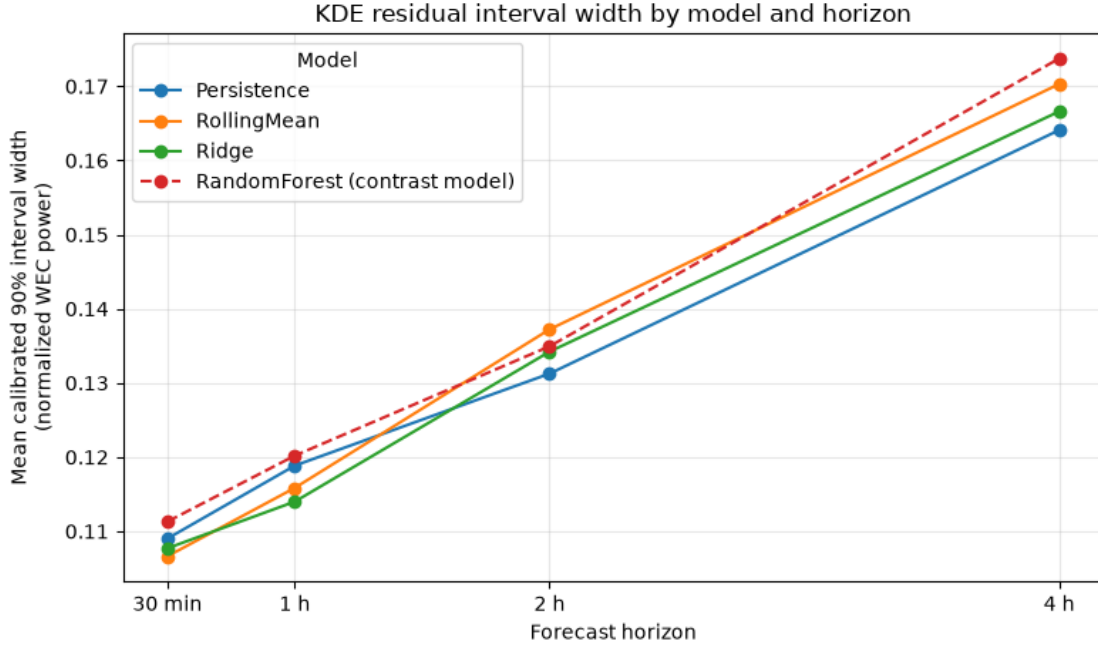
The method is still residual-based: it does not retrain the forecasting models or introduce new predictors. Instead, it fits a one-dimensional scikit-learn `KernelDensity` estimator to the calibration residuals, then extracts lower and upper residual quantiles from the smoothed distribution using a numerical CDF grid.

A fixed bandwidth of 0.01 normalized power is used for this lightweight comparison. This corresponds to smoothing over about 1% of the normalized WEC-power range, or about 2.5 kW on the 250 kW interpretation scale. The bandwidth is not tuned; it is chosen to keep the KDE comparison simple and avoid turning this notebook into a bandwidth-selection study.

Compared with direct residual quantiles, KDE can reduce the discreteness of empirical quantiles by smoothing the residual distribution. However, it still introduces a bandwidth choice and should therefore be interpreted as a lightweight distribution-smoothed comparison rather than a more

authoritative uncertainty method.

The simpler residual quantile and conformal-style intervals remain the main empirical interval baselines.



The KDE residual intervals show a similar horizon-dependent widening pattern to the residual quantile and conformal-style intervals. With the fixed bandwidth used here, KDE behaves as a reasonable smoothed-residual comparison rather than producing excessively wide intervals.

Because this method depends on a bandwidth choice, it remains secondary to the simpler empirical interval methods. Its main role is to check whether smoothing the calibration residual distribution changes the interval-width pattern in a meaningful way.

## 4. Prediction interval results and comparison

The interval rows generated above are now combined and evaluated on the test folds.

The main reliability metric is empirical coverage: the share of test observations where the realized estimated WEC power falls inside the prediction interval. Width-based metrics are also computed because very wide intervals can achieve high coverage without being practically useful.

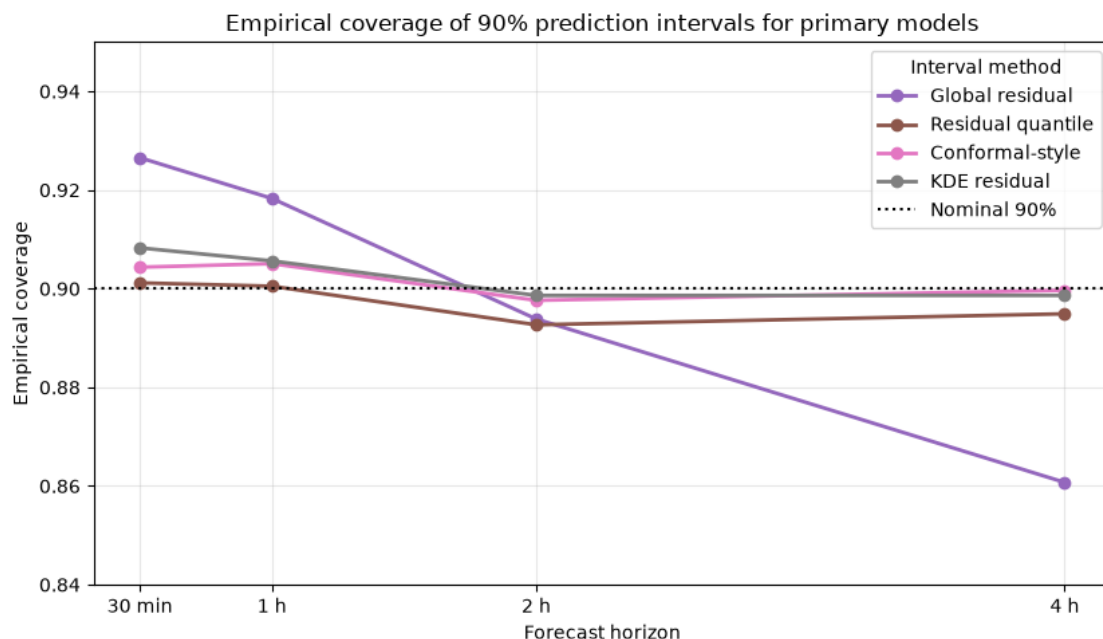
### 4.1 Empirical coverage by horizon

This section compares the empirical test-set coverage of the prediction intervals across forecast horizons.

For a nominal 90% interval, empirical coverage close to 0.90 indicates good calibration. Coverage below the nominal level indicates under-coverage, while coverage above the nominal level often means the interval is conservative or wider than necessary.



The coverage values are averaged across the primary models: Persistence, RollingMean, and Ridge. RandomForest is excluded from this summary plot because it is retained only as a contrast model.



The 90% prediction intervals achieve empirical coverage close to the nominal level for the primary models. The global residual baseline is slightly conservative at short horizons but under-covers at the 4 h horizon, which is consistent with its use of one pooled residual rule rather than horizon-specific calibration.

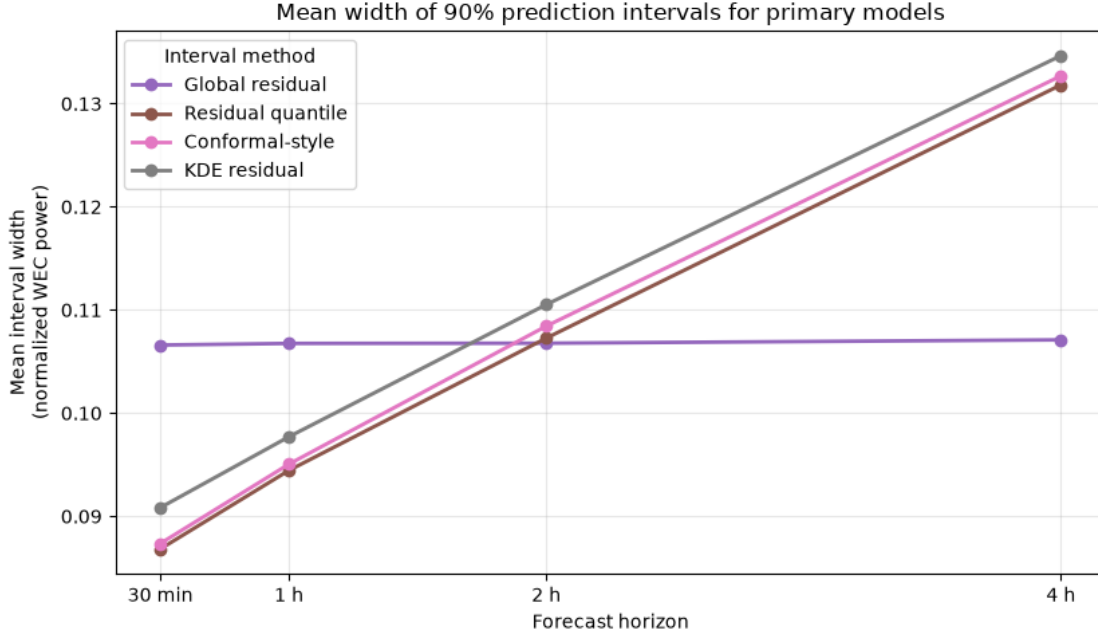
The residual quantile, conformal-style, and KDE residual methods stay closer to the nominal 90% level across horizons. This supports using model- and horizon-specific interval calibration for the main uncertainty analysis.

The y-axis is narrowed to highlight differences around the nominal 90% coverage level; the plotted values are averaged across the primary models: Persistence, RollingMean, and Ridge.

## 4.2 Interval width by horizon

Coverage alone is not sufficient, because very wide intervals can achieve high coverage without being practically useful.

This section compares the average interval width across forecast horizons for the primary models. Lower width is preferable only when empirical coverage remains close to the nominal level.



The interval-width comparison shows the trade-off behind the coverage results. The global residual baseline has nearly constant width across horizons because it uses one pooled residual rule, which helps explain why it is too conservative at short horizons and under-covers at the 4 h horizon.

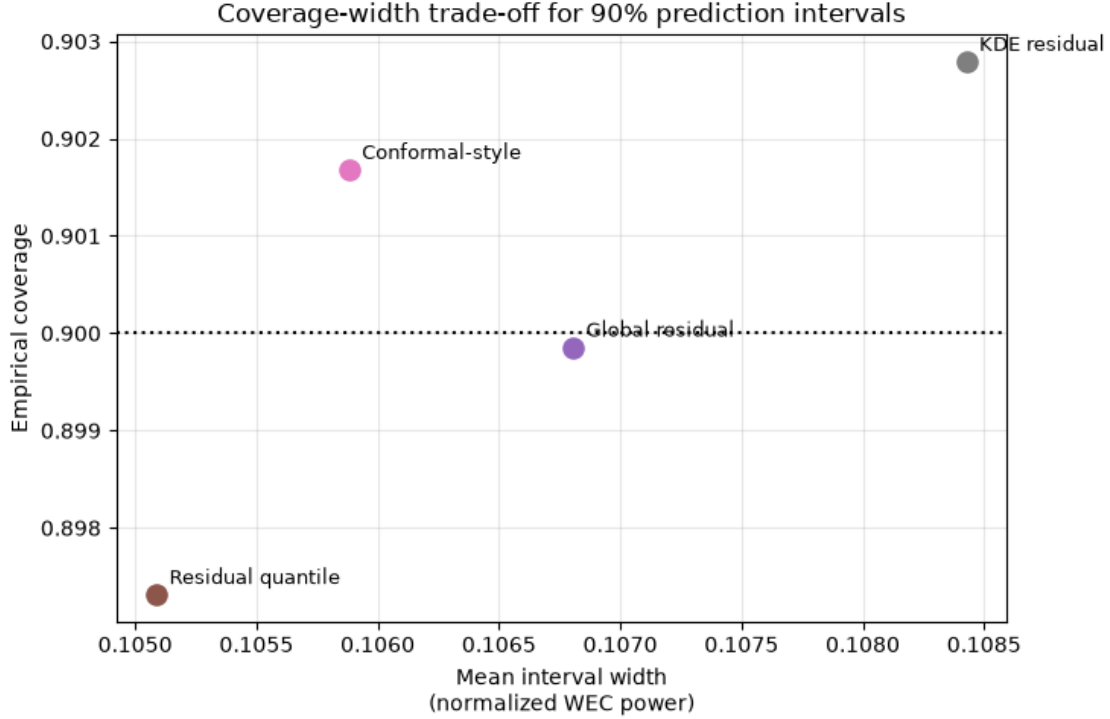
The model- and horizon-specific methods widen as the forecast horizon increases, matching the increasing forecast uncertainty seen in the residual diagnostics. Residual quantile intervals are slightly narrower than the conformal-style and KDE residual intervals, while KDE gives the widest intervals among the horizon-specific methods.

Overall, the horizon-specific methods provide more adaptive uncertainty estimates than the global baseline, while residual quantile intervals appear to offer the sharpest 90% intervals among the calibrated methods shown here.

### 4.3 Coverage-width trade-off

A useful prediction interval method should be both reliable and reasonably sharp.

This section compares empirical coverage against mean interval width for the 90% intervals. Points closer to the nominal coverage line with lower interval width are preferable.



The coverage-width trade-off shows that all four methods are close to the nominal 90% coverage level when averaged across primary models and horizons. The differences are therefore mainly in sharpness.

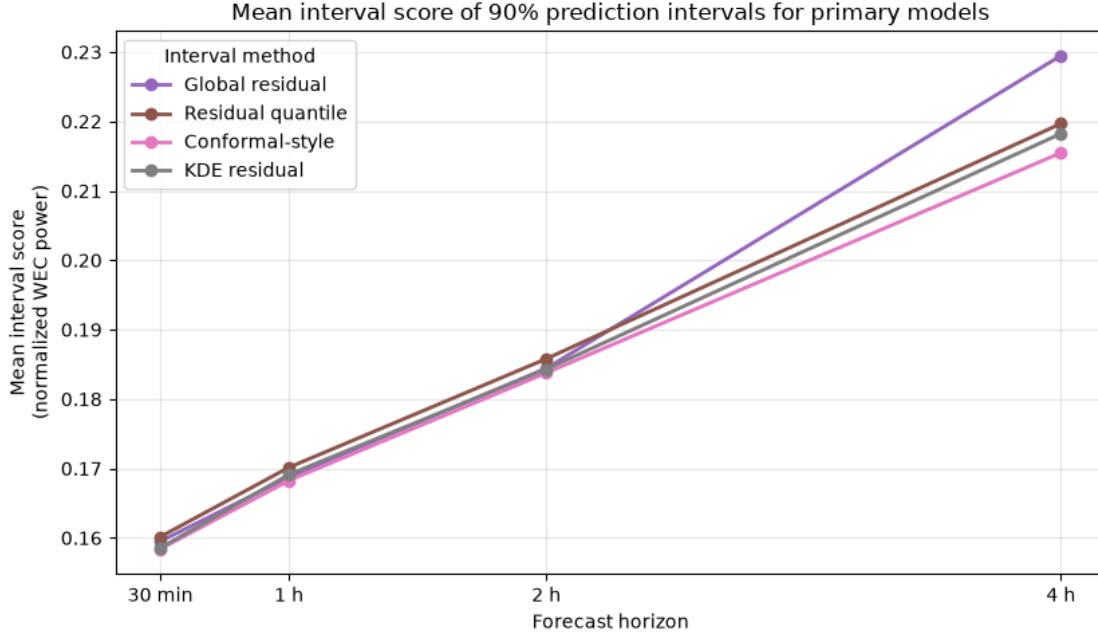
The residual quantile method gives the narrowest intervals, but with slightly lower average coverage. The conformal-style and KDE residual methods are somewhat wider and slightly more conservative. The global residual baseline sits close to the nominal level on average, but the horizon-wise results showed that this average hides under-coverage at the longest horizon.

Overall, the residual quantile and conformal-style methods appear to be the most useful main candidates: residual quantile intervals are sharper, while conformal-style intervals are slightly more conservative and easier to interpret as symmetric error bands.

#### 4.4 Interval score by horizon

The interval score combines interval width with a penalty when the realized value falls outside the interval. Lower values are better.

This metric is useful because it discourages intervals that are too wide, while also penalizing intervals that miss the true value.



The interval score increases with forecast horizon for all methods, reflecting the greater difficulty of longer-ahead WEC power forecasting.

Although the global residual baseline looked reasonable in average coverage, its interval score becomes clearly worse at the 4 h horizon. This is consistent with the earlier coverage and width results: one pooled interval rule is not flexible enough across horizons.

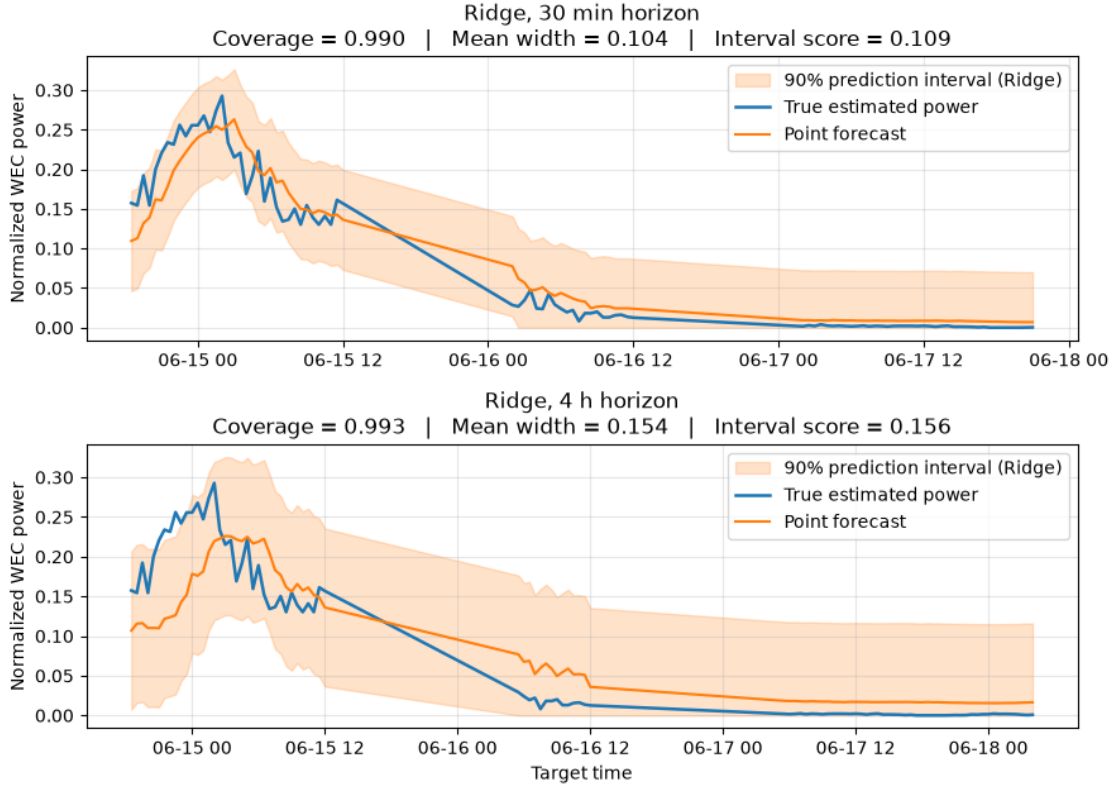
Among the horizon-specific methods, the conformal-style intervals have the lowest interval score at most horizons, while the residual quantile and KDE residual methods are close. This suggests that the conformal-style method provides a strong balance between coverage and sharpness for the main 90% interval analysis.

## 5. Example prediction intervals through time

The aggregate metrics compare reliability and sharpness, but they do not show how prediction intervals behave through time.

This section visualizes example 90% **conformal-style** intervals for **Ridge** forecasts at the shortest and longest forecast horizons. Residual quantile intervals were slightly sharper, but the conformal-style intervals gave a strong balance between coverage and interval score and are easier to interpret visually as symmetric calibrated error bands around the point forecast.

Example 90% conformal-style prediction intervals for estimated WEC power



The example intervals illustrate how the aggregate metrics translate into time-dependent uncertainty bands. The 30 min horizon has a narrower band and follows the observed estimated power more closely, while the 4 h horizon has a wider band and a smoother, more lagged point forecast.

Both example windows show high empirical coverage for the selected fold and method. However, the wider 4 h interval also has a larger mean width and higher interval score, reflecting the greater uncertainty and lower sharpness at longer forecast horizons.

This visual example supports the earlier metric-based result: conformal-style intervals provide reliable uncertainty bands, but the cost of maintaining coverage increases as the forecast horizon grows.

## 6. Physical interpretation on the 250 kW WEC scale

The main uncertainty evaluation uses normalized WEC power. For interpretation, the same interval widths can be converted to the 250 kW scale used for the generic WEC proxy.

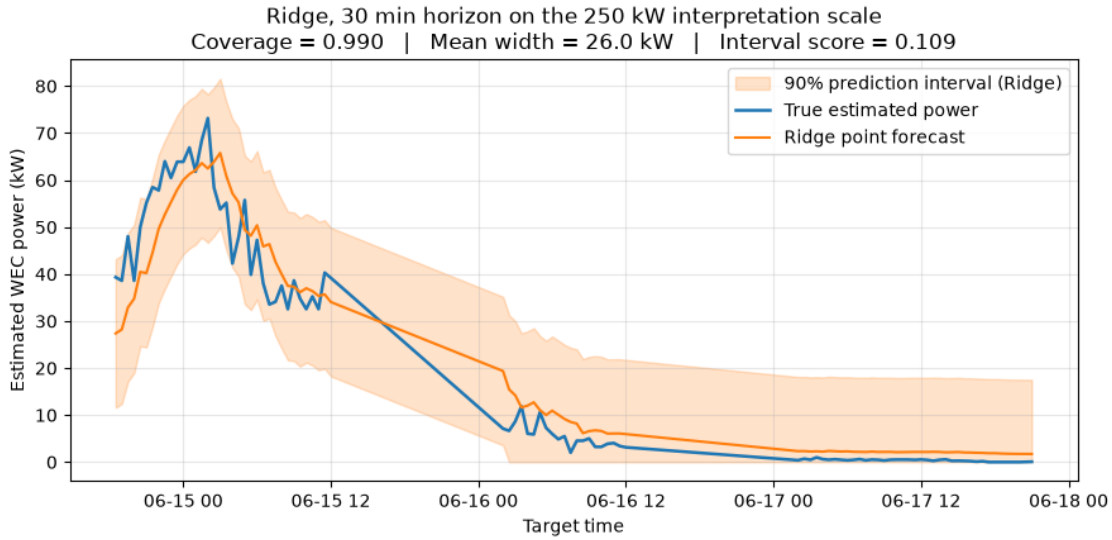
This conversion does not change the modelling results. It only expresses the normalized uncertainty bands in approximate kW terms for the simplified 250 kW device scale.

### Example 90% conformal-style intervals on the 250 kW interpretation scale

| Horizon | Coverage | Mean width<br>(norm.) | Mean width (kW,<br>250 kW scale) | Median width (kW,<br>250 kW scale) | Interval score<br>(norm.) |
|---------|----------|-----------------------|----------------------------------|------------------------------------|---------------------------|
| 30 min  | 0.99     | 0.104                 | 26                               | 26.8                               | 0.109                     |
| 4 h     | 0.993    | 0.154                 | 38.6                             | 37.5                               | 0.156                     |

### 6.1 Example interval on the 250 kW interpretation scale

The normalized interval can also be expressed on the 250 kW scale used for the generic WEC proxy. The plot below shows the same Ridge 30 min example window, now converted to approximate kW values.



The 250 kW interpretation scale makes the interval magnitude easier to read in physical terms. For the selected Ridge 30 min example, the 90% conformal-style interval has a mean width of about 26 kW, which corresponds to the same uncertainty band previously shown on the normalized scale.

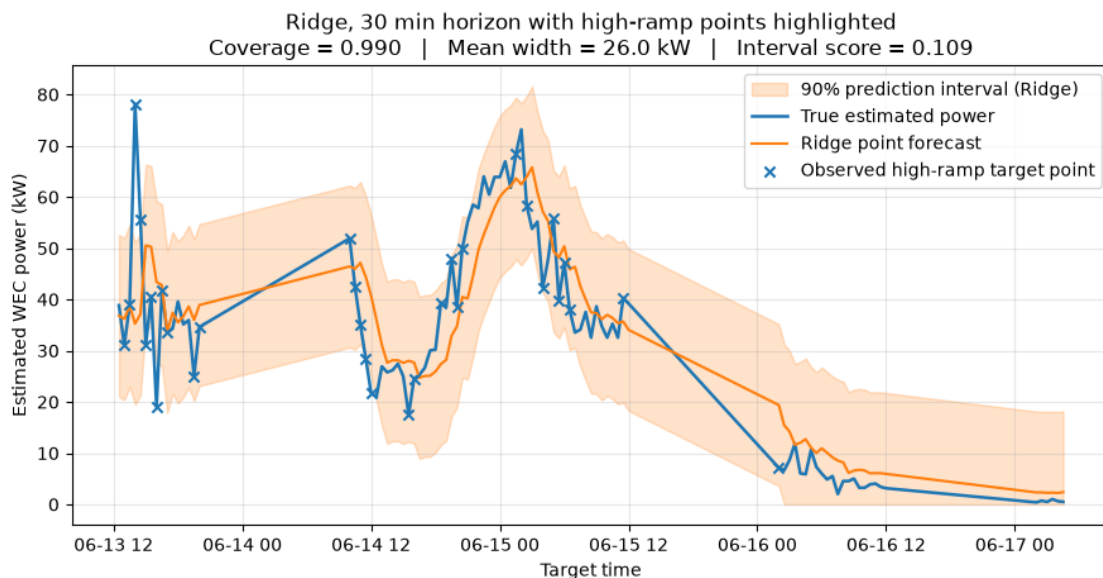
The small summary table also shows how this uncertainty grows with forecast horizon: the 4 h example interval is wider, at about 39 kW on average, compared with about 26 kW at 30 min. This is consistent with the broader interval bands and higher interval scores observed earlier for longer forecast horizons.

These kW-scale values provide a practical interpretation of forecast uncertainty for the generic 250 kW WEC proxy, but they are still not storage-sizing results. Their main role here is to connect the normalized interval analysis to a more intuitive device-scale context before the storage-aware smoothing analysis in the next notebook.

## 7. Uncertainty during high-ramp periods

As a lightweight bridge to the storage-aware smoothing notebook, this section checks how prediction intervals look during rapid changes in estimated WEC power.

High-ramp periods are marked using the largest observed target-to-target changes in the test data. This is only an exploratory visualization; detailed ramp and storage-smoothing metrics are left for Notebook 05.



The high-ramp markers are calculated from the observed estimated-power series, not from the Ridge forecast. They highlight target times where the realized 250 kW-scale estimated power changed sharply relative to the previous target value.

In this example, several high-ramp points are covered by the 90% Ridge prediction interval, but the point forecast remains smoother and sometimes lags the sharp observed changes. This illustrates why ramp periods are important for the next storage-aware notebook: even when the interval covers the realized value, rapid power changes can still be operationally relevant for smoothing and storage dispatch.

This section is therefore kept as a visual bridge rather than a full ramp-forecasting analysis. Detailed ramp-rate and storage-smoothing metrics are left for Notebook 05.

## 8. Summary and limitations

This final section summarizes the main 90% interval results across the primary models and forecast horizons.

### Summary of 90% interval methods across primary models and horizons

| Interval method | Mean interval score | Mean coverage | Mean coverage error | Mean width (norm.) | Mean width (kW, 250 kW scale) |
|-----------------|---------------------|---------------|---------------------|--------------------|-------------------------------|
| Conformal-style | 0.181               | 0.902         | 0.002               | 0.106              | 26.5                          |
| KDE residual    | 0.183               | 0.903         | 0.003               | 0.108              | 27.1                          |

| Interval method   | Mean interval score | Mean coverage | Mean coverage error | Mean width (norm.) | Mean width (kW, 250 kW scale) |
|-------------------|---------------------|---------------|---------------------|--------------------|-------------------------------|
| Residual quantile | 0.184               | 0.897         | -0.003              | 0.105              | 26.3                          |
| Global residual   | 0.186               | 0.9           | -0                  | 0.107              | 26.7                          |

The summary table is sorted by mean interval score, where lower values indicate a better balance between interval width and missed-coverage penalties. On average across the primary models and horizons, the conformal-style method gives the lowest interval score, followed closely by the KDE residual and residual quantile methods.

All four 90% interval methods achieve mean empirical coverage close to the nominal level. The average differences are small, so the ranking should not be interpreted as a strong universal winner. Instead, the horizon-wise figures provide the more important interpretation: model- and horizon-specific methods adapt better to the increasing forecast uncertainty than the global residual baseline.

The residual quantile method gives the sharpest intervals, while the conformal-style method provides a transparent symmetric error-band interpretation and the best average interval score in this comparison. KDE residual intervals are retained as a lightweight smoothed-residual comparison, but they remain secondary because they depend on a bandwidth choice.

The 250 kW-scale examples show that the normalized interval widths correspond to uncertainty bands of roughly 26–27 kW on average for the generic WEC proxy, with wider intervals at longer horizons in the example windows. The high-ramp visualization also shows that intervals may cover sharp target changes even when point forecasts remain smoother or lagged, motivating the storage-aware smoothing analysis in the next notebook.

Important limitations remain. The target is estimated WEC power, not measured WEC output. The intervals are calibrated from empirical residuals and should not be interpreted as formal physical uncertainty bounds. The conformal-style intervals are described cautiously because the data are time-dependent and the calibration block was also used for model selection in Notebook 03. Finally, the 250 kW-scale values are interpretation aids only; device scaling, array assumptions, and storage-smoothing metrics are left for Notebook 05.